



Department of Computer Science and Business Systems
Academic Year 2022 – 2023(Even Semester)

Degree, Semester & Branch: B.Tech., II & CSBS

Course Code & Title: AD3251 Data structures design

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Innovative Practice Description

- **Unit / Topic:** Unit 4 / Binary Search Tree and AVL Tree
- **Course Outcome:** CO 4 – Apply an efficient tree structures to perform searching, indexing and sorting
- **Topic Learning Outcome:** TLO 12 - Design and implement tree data structure.
- **Activity Chosen:** Visualization
- **Justification:**
 - Visualization is a teaching strategy which refers to the practical visualization of the tree data structure will improve the understanding of the students on the properties of the tree and its operations.
 - Need: Theoretical explanation of concepts is not sufficient and also not understandable one for beginners.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 1. **Material used :**
 - a. Virtual Lab - <https://ds1-iiith.vlabs.ac.in/Introduction.html>
 2. **Prerequisite :** Knowledge on the properties and operations on Binary Search Tree and AVL Tree. (Obtained by Lecture)
 3. Visualization of data structure is shown with the help of virtual lab.
 4. At each step, students are asked to perform on their own. The answers are verified in the virtual lab.
 5. Students also motivated to execute it for all data structures in their laptop or in their lab session.
- **CO – PO / PSO mapping:**

CO	PO1	PO3	PO12	PSO2
CO5	2	1	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO12	PSO2
	2	1	2	2
Justification for correlation	Apply the knowledge of engineering fundamentals	Design solutions for complex engineering problems	Recognize the need and have ability to engage in independent and life-long learning.	Exhibit the proficiency in software development

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students felt that Visualization on the topic gave them better understanding on the topic very easily. At Any time, students can verify their answers with the help of virtual lab.

- ❖ **Benefit of the practice:**

Flipped mode of teaching attracted the student's attention. They gave better focus on the topic than conventional teaching method.

- ❖ **Challenges faced in implementation:**

It is not a student centered method. So, few students not actively participate in this type of learning.

References:

1. <https://ds1-iiith.vlabs.ac.in/Introduction.html>